How Far Does a Frozen VGGT Go?
A Controlled Study of Feed-Forward 3D Instance Segmentation

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Abstract

Feed-forward 3D instance segmentation from unposed images is mostly built by adapting a geometry foundation model: recent methods fine-tune VGGT or Depth Anything 3, fully or with LoRA, and two of them report sixteen GPU-days of training. We study the alternative in which the backbone is never updated, and ask how much it gives up under the protocols of the adapted methods. A Mask DINO-style decoder whose queries are shared across views is trained on cached features of a frozen VGGT-1B (23 M trainable parameters, a few GPU-hours), and its masks are lifted to 3D with the backbone's own depth and cameras. Placing the result next to published ones required an audit, because published numbers differ along four axes that are not properties of the evaluator: how 2D masks reach the benchmark point cloud, the label setting, the number of views, and whether the evaluation domain was seen in training. We measure each axis with our own model; the first alone is worth a factor of 2.3 in AP_{50}. Trained on ScanNet and evaluated at about 50 views, as the published results are, the frozen model reaches 5.3 AP / 17.0 AP_{50} / 54.2 AP_{25} in the unposed, class-agnostic setting, above the zero-shot results of two adapted models. Trained without ScanNet it reaches 0.9 / 3.2 / 30.1 at 17 views: level with them at AP_{25} and three times lower at AP_{50}, which is also where PanSt3R, an earlier frozen-backbone method, stands. Against an adapted model trained on the same split as ours, a factor of 2.8 in AP_{50} remains once the transfer is equalised. Single-variable ablations show that cross-view mechanisms change 3D AP_{50} by −57% to +67%, and that a 2D mask metric ranks them differently and misses the largest gain; results on three further benchmarks place much of the remaining error in the lifting through frozen geometry. Code, protocols and the full measurement record are available at https://github.com/NicoloIacobone/vggt.

1 Introduction

Feed-forward reconstruction models such as DUSt3R [56], VGGT [54] and Depth Anything 3 [30] recover cameras, depth and point maps from a handful of unposed images in a single pass. A natural next question is whether the same networks can carry scene understanding, and 3D instance segmentation has become a test case. Three recent methods answer it in the same way. SegVGGT [43] inserts object queries into every layer of VGGT and adapts the backbone with LoRA [15]; FAST3DIS [25] adds 3D-anchored queries to a LoRA-adapted Depth Anything 3; IGGT [28] fine-tunes VGGT end to end on a curated multi-source dataset. All three change the backbone, and the two that report their cost use eight GPUs for two days.

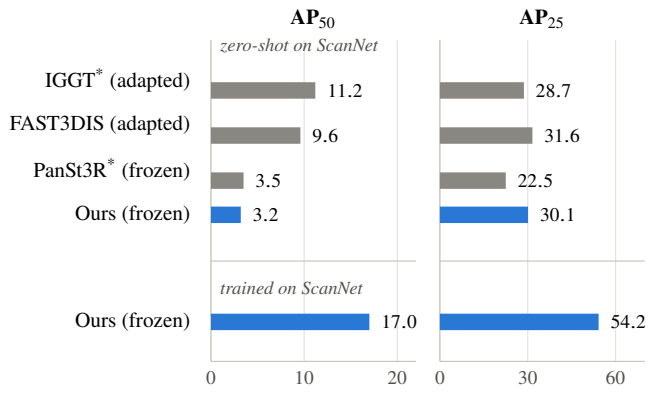


Figure 1: Adapted and frozen backbones on ScanNetv2 (unposed transfer, class-agnostic). Without ScanNet in training, the two frozen-backbone models sit at about a third of the AP_{50} of the two adapted ones, while at AP_{25} our frozen VGGT is level with them. With ScanNet in training the same recipe is far above all four, which is a data effect and not a ranking. Grey: published numbers at 50 views [25]; *re-evaluated there. Blue: this work, at 17 views (zero-shot) and 47 views (in-domain).

The alternative is to leave the backbone untouched and train only a decoder. It keeps the geometry outputs intact, lets the features be computed once and cached, and reduces training to a small head. PanSt3R [69] took this route for panoptic segmentation on a frozen MUSt3R [4] and reports that fine-tuning the backbone changed little. How much a frozen backbone gives up for 3D instance segmentation, measured under the protocols of the adapted methods, is not known. Answering that runs into a second problem: there is no common ruler. On the ScanNet benchmark [10], with the same official evaluation code, SegVGGT reports 71.7 AP_{50} and FAST3DIS reports 9.6. A sevenfold gap between two contemporaneous methods of the same family is a sign that the two numbers do not measure the same thing.

This paper addresses both problems. We attach a Mask DINO decoder [27] to a strictly frozen VGGT-1B. Its queries are shared by all views of a scene, so that one query is one instance in every view by construction and no mask matching or fusion stage is needed; a cross-frame attention block ties the copies of a query together, and the usual 2D anchor boxes can be replaced by 3D anchors read off the backbone's point head.

The head has 23 M parameters and trains in about four hours on one GPU. We then evaluate it under every protocol used by the methods above: with predicted and with ground-truth geometry in the 2D-to-3D transfer, with and without class labels, at 17 and at about 50 views, and with and without ScanNet in the training set.

Findings. *(i) The published numbers are not mutually comparable.* SegVGGT transfers its masks to the benchmark point cloud with ground-truth poses and sensor depth, while FAST3DIS and IGGT are scored with predicted geometry. On identical masks of ours the two transfers differ by a factor of 2.3 in AP_{50}, consistently across five checkpoints (Section 5.3). *(ii) The frozen model leads only where it has a data advantage.* Trained on ScanNet and evaluated at about 50 views, it reaches 5.3 / 17.0 / 54.2 (AP / AP_{50} / AP_{25}), above the zero-shot numbers that FAST3DIS and IGGT report at 50 views. With ScanNet removed and the composition of IGGT's training mixture reproduced, it reaches 0.9 / 3.2 / 30.1 at 17 views: on par at AP_{25}, about three times lower at AP_{50}, and next to PanSt3R's 0.8 / 3.5 / 22.5 (Figure 1). Against SegVGGT, which trains on the same split as we do, a factor of 2.8 remains after the transfer is equalised (Sections 5.2 and 5.3). *(iii) What matters for the 3D metric is poorly predicted in 2D.* Removing cross-frame attention costs 57% of the 3D AP_{50}; 3D anchors add 67% while the 2D bundle metric stays within seed noise; and a 2D metric ranks the mechanisms in a different order from the 3D one (Section 5.4). A cross-view identity explanation that we initially held for the latter is not supported by standard tracking metrics, and we report that as well (Section 5.5). *(iv) The lifting limits the result more than the mask decoder does.* More views stop helping at about 50, and on ScanNet++ and Replica the unposed result collapses to zero while the posed one survives and improves with training data (Sections 5.6 and 5.8).

Contributions.

- A controlled study of 3D instance segmentation on a strictly frozen VGGT, with single-variable ablations of the cross-view mechanisms scored on the official 3D benchmark.
- An audit of comparability: four axes on which published numbers differ, each quantified with the same model, and an oracle that certifies our implementation of the posed transfer.
- Training-matched comparisons that place an in-domain result next to its zero-shot counterpart, and that put the distance between frozen and adapted backbones at about a factor of three in AP_{50}, narrowing at AP_{25} (to parity in the zero-shot case), under confounds that we state.
- Evidence, across four benchmarks and seven training mixtures, on where the remaining error sits.

We do not claim a new state of the art. The contribution is the set of measurements and the protocol that makes them comparable; the code and the complete record of runs, including the failed ones, are public.

2 Related Work

Feed-forward 3D reconstruction. DUSt3R [56] and MASt3R [24] regress point maps from image pairs; MUSt3R [4], CUT3R [55] and Fast3R [59] extend the idea to many views. VGGT [54] alternates per-frame and global attention over DINOv2 tokens [41] and predicts cameras, depth, point maps and tracks in one pass; π^3 [57], Depth Anything 3 [30] and MapAnything [19] follow. These backbones are now used as a substrate for detection, occupancy and driving [5, 8, 18], so that a decoder on top of VGGT is a pattern and not by itself a contribution.

3D instance segmentation. The strongest ScanNet results come from models that consume a reconstructed point cloud, such as Mask3D [49], OneFormer3D [22] and Relation3D [34], or posed RGB-D frames, alone or together with the point cloud, such as ODIN [16] and SegDINO3D [42]. A second family lifts 2D masks of a foundation model [21, 45] onto known geometry [40, 52, 58, 60, 63]. A third obtains multi-view consistency by optimising a per-scene radiance field or Gaussian representation [3, 20, 37, 50, 61]. All of them need a depth sensor, camera poses, or minutes of optimisation per scene. The setting studied here needs none: the input is a set of unposed RGB images and the output is produced in one forward pass.

Segmentation on feed-forward geometry backbones. LSM [13] and SAB3R [7] add semantic features to pairwise reconstruction models. PanSt3R [69] is the closest predecessor in design: it keeps MUSt3R and DINOv2 frozen and trains a Mask2Former-style decoder [9] whose queries are shared by all views, for panoptic segmentation, evaluated by panoptic quality. EPS3D [67] targets open-vocabulary panoptic segmentation by distillation. For class-agnostic or closed-set 3D instance segmentation, IGGT [28] fine-tunes VGGT with an instance-contrastive objective and clusters the resulting features, SegVGGT [43] places object queries inside all aggregator layers of a LoRA-adapted VGGT, and FAST3DIS [25] anchors queries in 3D on a LoRA-adapted Depth Anything 3. Related lines address online, promptable or cross-view mask association [11, 14, 17, 65]. We do not claim the two mechanisms we ablate: view-shared queries appear in PanSt3R and SegVGGT, and 3D-anchored queries in FAST3DIS. What has not been available is a controlled comparison in which one backbone, one decoder and one protocol are held fixed while these mechanisms, the training data and the evaluation protocol are varied one at a time, and in which the backbone is frozen.

Query-based detection and segmentation. Our decoder is Mask DINO [27], which combines the mask branch of Mask2Former [9] with the DETR line of detectors [6]: deformable attention [68], anchor-box queries [32], denoising training [26] and their combination in DINO [64]. The feature pyramid follows the single-scale ViT recipe of ViTDet [29].

Frozen backbones and evaluation audits. Probing studies ask what frozen visual foundation models encode about 3D

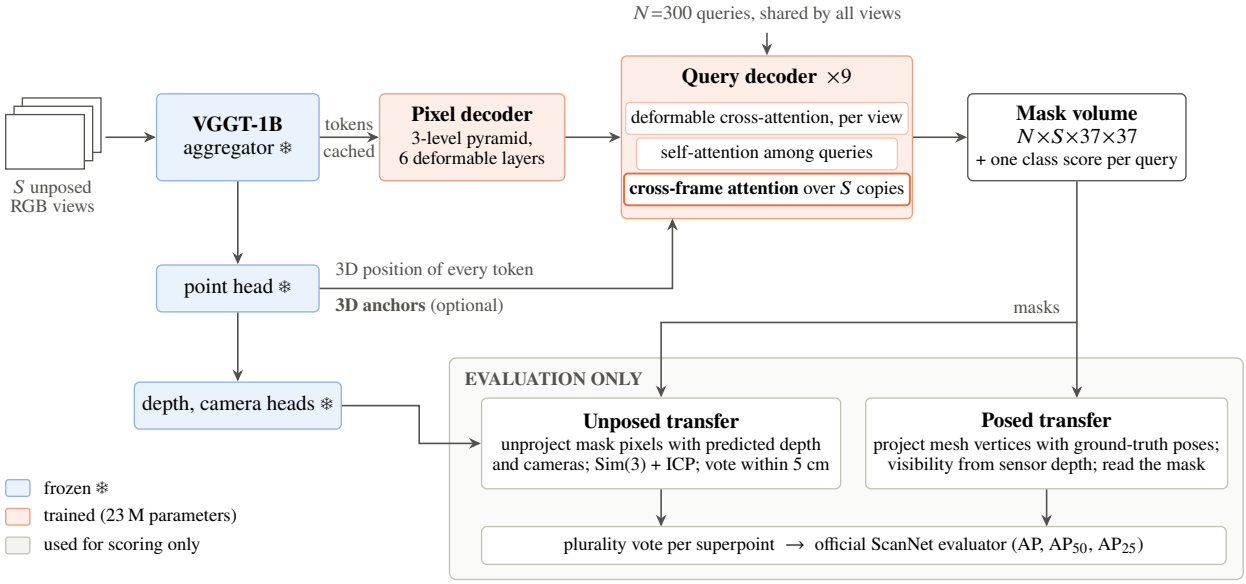


Figure 2: Model and evaluation. A frozen VGGT-1B turns S unposed views into tokens that are cached once per scene. The trainable head is a Mask DINO pixel decoder and query decoder; its N queries are shared by all views and their per-view copies are tied by cross-frame attention, so the output is one mask volume per query. Optionally the anchor of a query is a 3D point taken from the frozen point head. For scoring only, the masks are carried to the benchmark point cloud in one of two ways: with the backbone's own predicted geometry (unposed) or with ground-truth poses and sensor depth (posed).

structure [12, 36]; we ask the converse question of a geometry model, namely what a frozen one supports for instance-level understanding. In methodology, the paper is closest to work that re-examines reported progress under matched conditions [39].

3 A Decoder on Frozen Geometry Features

Figure 2 gives an overview. The design goal is to change as little as possible: the backbone is used as released, the decoder is a faithful port of a standard one, and the two additions that make it multi-view are small and can be switched off independently.

3.1 Frozen backbone and feature cache

VGGT [54] embeds each of S input views with DINOv2 and passes the tokens through 24 blocks that alternate attention within a frame and attention across all frames. We read the tokens of the last block. With 518×518 inputs and a patch size of 14, each view yields a 37×37 grid of 2048-dimensional tokens; the camera and register tokens are discarded. The backbone stays in inference mode and receives no gradient. Its tokens are computed once per scene before training and stored, so an optimisation step involves the head only.

How the tokens are computed is an experimental variable. By default all views of a training sample pass through the backbone together, so the tokens of one view already depend on the other views through global attention (*multi-view features*). The alternative runs the backbone on each view alone (*per-frame features*).

3.2 Per-view mask decoder

The head is Mask DINO [27], ported without changes to its logic.

Pixel decoder. VGGT provides a single feature resolution. Following ViTDet [29], a 1×1 convolution maps the token grid to 256 channels at 37×37, and two stride-2 convolutions add levels at 19×19 and 10×10. Six multi-scale deformable attention layers [68] refine the three levels jointly. The finest level, after a further 1×1 convolution, serves as the mask feature map $\mathbf{F} \in \mathbb{R}^{256\times37\times37}$.

Query decoder. N=300 queries are initialised by two-stage selection: the encoder tokens are classified, the N highest-scoring ones provide the initial content embeddings, and their initial masks are converted to boxes that serve as anchors [32]. Nine decoder layers apply self-attention among queries, deformable cross-attention around each query's anchor, and a feed-forward block, and refine the anchor after every layer. A query q predicts class logits, a box, and a mask obtained as the inner product of an embedding of q with $\mathbf{F}$, at 37×37 resolution. Training uses Hungarian matching [23], a focal classification loss [31], binary cross-entropy and Dice mask losses [38], L_1 and GIoU box losses [46], denoising queries [26, 64] and deep supervision on every layer. Classification uses sigmoid logits without a background column, so "no object" is a query whose logits are all low.

3.3 Queries shared across views

Applied frame by frame, the decoder above produces unrelated query sets, and 3D instances would have to be recovered by matching masks across views afterwards. We instead let one

query set serve all views of a *bundle* of S frames, with three changes.

Shared initialisation. Two-stage selection takes the top N proposals from the union of the encoder tokens of all S views, and the selected content embedding is copied to every view. Each view keeps its own anchor box, derived from that view's initial mask and refined per view. Content is shared; image-plane geometry is per view.

Cross-frame attention. After each decoder layer, the S copies of a query attend to one another in a standard self-attention block applied along the view dimension. The block has no positional encoding of the view index: a bundle is an unordered set, and the block therefore runs unchanged at test time on a different number of views than it was trained with. Denoising queries are excluded.

Bundle-level matching. Ground-truth instances carry an identity that is consistent across views, because the 2D annotations are projections of one 3D annotation. The Hungarian assignment is computed once per bundle: the classification cost uses the mean score over views, the mask costs are evaluated on the concatenated $S{\times}37{\times}37$ volume, and the box costs are averaged over the views in which the instance is visible. The assignment is then applied per view with the unchanged single-view losses. In a view where its instance is not visible, a matched query is supervised as "no object".

The result is a mask volume $\mathbf{M} \in \mathbb{R}^{N\times S\times 37\times 37}$ in which index q denotes the same instance in every view, and one class score per query (the maximum over views). No matching, tracking or fusion step follows.

3.4 3D anchors

An anchor box lives in one image. For a query that represents an object seen from S viewpoints, a natural alternative is a single anchor in 3D, as in FAST3DIS [25]. We implement it inside the same decoder so that the two can be compared with everything else fixed.

VGGT's frozen point head predicts a 3D point for every pixel in a common frame. We average these points, weighted by their predicted confidence, over the 14×14 pixels of each token, and normalise the resulting token positions $\mathbf{p}$ per bundle to zero mean and unit root-mean-square radius. A query owns an anchor $(\mathbf{a}, \log r)$ with $\mathbf{a} \in \mathbb{R}^3$, initialised at the position of its selected token. The anchor is projected into view v as a soft assignment to that view's tokens,

$$w_j^v = \underset{j}{\mathrm{softmax}}\Big(-\|\mathbf{p}_j^v - \mathbf{a}\|^2/r^2\Big), \qquad \mathbf{c}^v = \textstyle\sum_j w_j^v\,\mathbf{u}_j, \quad (1)$$

where $\mathbf{u}_j$ is the image position of token j. The reference box of the query in view v is centred at $\mathbf{c}^v$ and its size is twice the standard deviation of $\mathbf{u}$ under w^v, floored at one token. It replaces the anchor box in the positional encoding and in the deformable sampling; each layer predicts an update of $(\mathbf{a}, \log r)$, averaged over views. The projection needs no camera parameters and is differentiable in the anchor. Losses and matching are unchanged, and 2D boxes are still predicted and supervised in every view.

3.5 Training and cost

One optimisation step processes one bundle of S=8 views. We use AdamW [33] with a learning rate of 10^{-4}, cosine decay and the loss weights of Mask DINO; details are in Appendix A. The head has 22.8 M parameters (23.0 M with 3D anchors), about 2% of the backbone. On the 1,201 ScanNet training scenes, the 28.8k steps of our default schedule take about four hours on one RTX 4090, after a single pass of the backbone over the data to fill the cache; an 84k-step schedule on a 3,520-scene mixture takes under 19 hours on one A100, caching included.

4 What Makes Two Numbers Comparable

The methods we compare to are scored on the same benchmark with the same evaluation code, and their numbers still differ in what they measure. Table 1 lists the axes. This section defines each one and states how we match it.

4.1 Benchmark and evaluator

We use the ScanNet v2 3D instance segmentation benchmark [10] on the official validation split (312 scans). A prediction is a set of vertex masks on the benchmark mesh with a class and a confidence. The official script reports AP, averaged over IoU thresholds from 0.5 to 0.95, and AP_{50} and AP_{25} at fixed thresholds, as a mean over 18 classes. We use a line-by-line port of that script. Feeding the ground truth of every validation scene back as predictions returns exactly 100 on all three metrics.

4.2 Two ways to carry a mask into 3D

An image-based model predicts masks in the views; the benchmark wants them on the mesh. Two procedures are in use, and they differ in whether the model's own geometry takes part.

Unposed transfer. This is the setting of FAST3DIS [25] and of its re-evaluation of IGGT and PanSt3R (predicted geometry, a Sim(3) alignment of the trajectory refined by ICP, and nearest-vertex correspondence), and it is our default. All sampled views of a scene are processed in one forward pass. Queries are ranked by class score and the top 100 are kept. Mask logits are upsampled to the input resolution and every pixel is given to its most probable query if that probability exceeds 0.5. The labelled pixels are unprojected with VGGT's predicted depth and cameras. Since this reconstruction has an arbitrary frame and scale, it is registered to the benchmark mesh by a similarity transform estimated from predicted and ground-truth camera centres [53] and refined by ICP [2]. Each point then votes for its query on the nearest mesh vertex within 5 cm, and every segment of the benchmark's over-segmentation is assigned to the query with the most votes, as in SegVGGT. The 5 cm radius is our choice and is left at this value throughout; its effect is measured in Section 5.6. Ground-truth poses enter only in the registration, that is, after the prediction is complete.

Table 1: Axes on which published feed-forward 3D instance segmentation numbers differ. None of them is a property of the evaluator: every row is scored with the official ScanNet evaluation code. "Posed" transfers finished 2D masks to the benchmark point cloud with ground-truth poses and sensor depth; "unposed" uses the model's own predicted geometry followed by a Sim(3) and ICP registration. Labels are class-aware or class-agnostic; "stride 20" means every 20th frame of the scan. *IGGT and PanSt3R report no 3D instance AP themselves; the protocol listed is that of their re-evaluation by FAST3DIS [25]. Our compute covers the runs whose duration was recorded; none was recorded for the models trained without ScanNet (Appendix A).

Method	Backbone	Training data	ScanNet in training	Mask-to-3D transfer	Labels	Views per scene	Training compute
SegVGGT [43]	VGGT + LoRA	ScanNet (1,201 scenes)	yes	posed	aware	stride 20	8 GPUs, 2 days
FAST3DIS [25]	DA3 + LoRA	ASE (40% of >100k scenes)	no	unposed	agnostic	50	not reported
IGGT [28]	VGGT, fine-tuned	InsScene-15K: ASE, Infinigen, RE10K, ScanNet++	no	unposed*	agnostic*	50*	8 GPUs, 2 days
PanSt3R [69]	MUSt3R, frozen	ScanNet++, ASE, Infinigen (2,541 scenes); COCO, ADE20K	no	unposed*	agnostic*	50*	not reported
Ours	VGGT, frozen	ScanNet, or mixtures with and without it	both	both	both	17 and 47	1 GPU, 4 to 19 h

Posed transfer. SegVGGT [43] proceeds in the opposite direction: every vertex of the ground-truth point cloud is projected into every view with the ground-truth pose and intrinsics, its visibility is decided by the sensor depth map, and the predicted mask is read at the pixel it lands on. Their paper states that ground-truth depth and poses are used in this mapping "for fair comparison". No predicted geometry is involved, and neither is registration. We implement the same procedure, followed by the same vote over segments.

What each one measures. The posed transfer measures the quality of the multi-view masks. The unposed transfer measures the product of mask quality and reconstruction quality. Both are legitimate, and both describe models that take only unposed images as input; what cannot be done is to read a number obtained with one against a number obtained with the other. We report both for our models, in separate columns.

To make sure that our posed transfer reads masks at the right pixels, we render the ground-truth 3D instances into every view through it and feed the result back as predictions. Of the annotated vertices that receive an instance this way, 99.99% receive their own, and the oracle scores 82.8 / 94.8 / 97.4 at the 17 views per scene of the standard ScanNet frame export. The shortfall from 100 is coverage: about 9% of annotated vertices are seen by none of those views.

4.3 Labels, views and training domain

Label setting. SegVGGT reports the class-aware benchmark metric. FAST3DIS ignores class labels, and so does its re-evaluation of IGGT and PanSt3R. We compute both from the same predictions, the class-agnostic one by collapsing all benchmark classes on both sides. Our class head is trained on 19 ScanNet categories: the benchmark's classes without *other furniture*, which the 2D supervision treats as background, plus wall and floor, whose predictions are dropped at evaluation. The class-aware mean therefore includes one class on which the model cannot score.

View budget. FAST3DIS samples 50 views uniformly per scene; SegVGGT takes every 20th frame. The frame export distributed with ScanNet contains every 100th frame, 17.4 per validation scan. We report at 17.4 views and, from a denser export of the same scans, at 50 requested views, which gives 46.7 on average because a fifth of the scans are too short.

Training domain. SegVGGT trains on the ScanNet training split. FAST3DIS, IGGT and PanSt3R do not train on ScanNet, so their ScanNet numbers are zero-shot. A model trained on ScanNet and evaluated on ScanNet is not comparable to them, however well the other axes are matched. We therefore train both ways (Section 5.2).

What remains unmatched. We can match the *composition* of a competitor's training set but not its scale or its compute, and we did not retrain or re-run any competitor. FAST3DIS does not specify which scenes it evaluates. These limits are restated where they matter.

5 Experiments

5.1 Setup

Data. The default training set is the official ScanNet v2 training split (1,201 scans) with the official 2D instance annotations, which are renderings of the 3D annotation and therefore carry view-consistent instance identities. For each scan we use about 100 frames, every fifth of the first 500 raw frames, and draw two bundles of eight: one evenly spaced and one random, the latter with brightness and contrast jitter. Evaluation uses the 312 validation scans. Further training sources and benchmarks are introduced in Sections 5.7 and 5.8 and listed in Appendix D.

Models. The *reference* configuration has view-shared queries, multi-view backbone features, cross-frame attention, 2D anchor boxes and a 19-class head. The configuration we carry forward adds 3D anchors. The models of the training-data study (Sections 5.7 and 5.8), its ScanNet-only baseline included, use a one-class head, since the other datasets do not share ScanNet's label set, and one evenly spaced bundle per scene.

Table 2: ScanNetv2 validation, unposed transfer, class-agnostic, grouped by whether ScanNet was seen in training. The two groups answer different questions and are not ranked against each other. Views are the mean number of input frames per scene (requesting 50 yields 46.7 because some scans are short). The 17.4-view in-domain row gives the range over two training seeds; all other rows of ours are single runs. *As re-evaluated by FAST3DIS [25]. †RE10K is supervised with SAM 2 pseudo-labels, as in InsScene-15K.

Method	Backbone	Training data	Views	AP	AP_{50}	AP_{25}
Zero-shot on ScanNet (ScanNet not seen in training)						
IGGT* [28]	VGGT, fine-tuned	InsScene-15K	50	2.8	11.2	28.7
FAST3DIS [25]	DA3 + LoRA	ASE (40%)	50	3.8	9.6	31.6
PanSt3R* [69]	MUSt3R, frozen	ScanNet++, ASE, Infinigen, COCO, ADE20K	50	0.8	3.5	22.5
Ours	VGGT, frozen	ScanNet++, Infinigen, RE10K†, ASE (4,819 scenes)	17.4	0.9	3.2	30.1
In-domain (trained on the ScanNet training split)						
Ours	VGGT, frozen	ScanNet (1,201 scenes)	17.4	3.9–4.2	12.9–13.8	48.5–50.4
			46.7	5.3	17.0	54.2
Ours + extra data	VGGT, frozen	ScanNet, ScanNet++, Infinigen (3,520 scenes)	17.4	5.7	16.6	51.6
			46.7	6.9	19.3	56.0

Hyper-parameters are given in Appendix A.

Selection, seeds and internal metrics. During training we monitor two 2D metrics computed with our own code on the 37×37 mask grid: AP_{50} per frame, and AP_{50} per bundle, which scores the mask volume of a query over eight views against the ground-truth volume. They are used to select the checkpoint (highest bundle AP_{50} on the validation split) and to compare with the 3D benchmark in Section 5.4; they are never placed next to published numbers. The parameters of the 3D lifting are fixed at their defaults and not tuned. The reference and the 3D-anchor models were each trained with two seeds: the spread is 0.9 points on 2D bundle AP_{50}, 0.9 points on 3D class-agnostic AP_{50}, 0.3 points on AP and 1.8 to 4.5 points on AP_{25}. Differences below this size are not interpreted. All other cells are single runs.

5.2 In-domain and zero-shot: two readings of one model

Table 2 compares with the published unposed, class-agnostic results. The ScanNet-only model has the 19-class head and is scored with its labels collapsed; the other models of ours in the table have a one-class head.

In-domain. Trained on ScanNet and evaluated at the 17.4 views of the standard frame export, the frozen model scores 3.9 to 4.2 AP, 12.9 to 13.8 AP_{50} and 48.5 to 50.4 AP_{25} over two seeds. At the view budget of the published rows it reaches 5.3 / 17.0 / 54.2, which is 1.4, 1.8 and 1.7 times the corresponding numbers of FAST3DIS and above IGGT on all three. With ScanNet++ and Infinigen added to the training set it reaches 6.9 / 19.3 / 56.0.

This is not a like-for-like result. The published methods have never seen ScanNet, and ours was trained on it. What the row shows is that a small head trained for a few hours on in-domain annotations gives better 3D instances than general models applied zero-shot, which is useful to know and unsurprising.

Zero-shot. To remove the advantage we train the same recipe without ScanNet, on the sources that make up IGGT's InsScene-15K: ScanNet++ [62], Infinigen [44] and RE10K [66] from the InsScene-15K release, and ASE [1] from its own public release, 4,819 scenes in total. The result is 0.9 / 3.2 / 30.1. At AP_{25} this is within 1.5 points of FAST3DIS and above IGGT, at a third of their view budget. At AP_{50} it is 3.0 and 3.5 times lower. Removing ScanNet from training costs the model a factor of about four in AP_{50} (13.8 to 3.2 at equal views, or 13.0 to 3.2 against the one-class ScanNet model of Table 6, which shares this recipe) and 1.6 to 1.7 in AP_{25}: most of the in-domain lead was the training data.

The zero-shot row is close to PanSt3R's 0.8 / 3.5 / 22.5, the other model in the table whose backbone is frozen, which was trained on an overlapping set of sources. Two independent frozen-backbone systems, on different backbones, thus land near 3 AP_{50}, against 10 to 11 for the two adapted ones.

What this does not establish. It does not show that freezing the backbone is the cause. Three differences remain in the zero-shot comparison. *Scale:* we use 1,000 ASE scenes where FAST3DIS uses 40% of more than 100,000, and InsScene-15K is, by its name, about three times larger than our mixture; its per-source composition is not reported. *Views:* our zero-shot model was scored at 17.4 views, and the in-domain model gained 24% in AP_{50} from 17 to 47 views (Section 5.6), which would narrow a threefold gap without closing it. *Design:* mask resolution, query count and the way instances are formed differ across all methods. The supportable statement is that, with training data of matched composition, the frozen model reaches the adapted ones at the coarse localisation threshold and not at the strict one.

5.3 Same training split, same transfer

SegVGGT is the one method that trains on the split we train on, so no data advantage exists in either direction. It is scored with the posed transfer and class-aware labels, and Table 3 reports our models in that setting next to the unposed one.

The transfer is worth a factor of 2.3. Swapping predicted for ground-truth geometry in the transfer, with masks, frames, queries and evaluator unchanged, raises AP_{50} from 6.7 to 15.6 for the reference model and from 11.2 to 25.7 with 3D anchors.

Table 3: The same masks under both transfers (ScanNetv2 val, class-aware, 17.4 views, default lifting parameters). Only the way a finished 2D mask reaches the point cloud changes between the two column groups. S is the number of views per training bundle. The oracle row pushes ground-truth masks through the posed transfer.

	Unposed			Posed		
Checkpoint	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
2D anchors, S=8	2.3	6.7	26.8	6.0	15.6	40.8
3D anchors, S=8	3.8	11.2	36.0	10.4	25.7	50.4
2D anchors, S=16	3.3	9.8	33.6	8.3	21.6	48.8
trained longer	3.2	11.5	41.4	8.8	26.0	57.2
3D anchors, S=16	3.2	10.9	35.3	8.2	23.6	50.1
GT masks (oracle)				82.8	94.8	97.4
SegVGGT [43]				50.4	71.7	87.0

The ratio lies between 2.2 and 2.3 for all five checkpoints. Coverage accounts for much of it: the fraction of mesh vertices that receive a vote rises from 15% to 34%, and the fraction of annotated vertices that end up in a predicted instance from 64% to 79%. A number obtained with the posed transfer and one obtained with the unposed transfer therefore differ by more than a factor of two before any property of the model enters.

A factor of 2.8 remains. In SegVGGT's own setting our best models reach 10.4 / 25.7 / 50.4 (3D anchors) and 8.8 / 26.0 / 57.2 (wider bundles, longer schedule), against 50.4 / 71.7 / 87.0. The raw ratio between their published AP_{50} and our unposed AP_{50} (71.7 to 11.2, a factor of 6.4) thus decomposes into 2.3 for the transfer and 2.8 that is not protocol. At AP_{25} the remaining distance is smaller, a factor of 1.5 to 1.7. The remaining differences all favour SegVGGT and we cannot separate them: a LoRA-adapted backbone with queries in all 24 layers, every 20th frame as input where we use every 100th, and masks at half the input resolution where ours are 37×37. One candidate is ruled out: keeping 600 instead of 100 queries per scene changes our class-agnostic AP_{50} from 13.8 to 14.0. The oracle row shows that the 17-view budget by itself caps the posed protocol at 94.8, far above where any model is. For orientation, SegVGGT's table also lists a reconstruct-then-segment baseline, OneFormer3D applied to a VGGT reconstruction, at 5.4 / 10.2 / 17.4.

5.4 Which mechanisms matter in 3D

Table 4 varies one training flag at a time on the official split and reports both the internal 2D metrics and the 3D benchmark. Figure 3 shows the two side by side.

Cross-frame attention. Removing the block that connects the per-view copies of a query, while keeping shared initialisation and bundle-level matching, lowers 3D AP_{50} from 6.7 to 2.9 (−57%). The loss is the same in the class-agnostic column (5.0 to 2.1), so it is not an artefact of the label setting. In 2D the same change costs 26% of the bundle metric and 10% of the per-frame one: objects are still found in each view, and what degrades is whether one query covers an object in all of them.

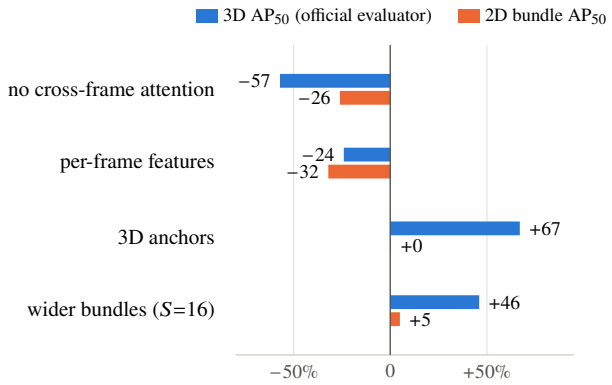


Figure 3: The same four changes, seen by a 2D and by the 3D metric. Relative change of AP_{50} with respect to the reference configuration (Table 4; 3D: class-aware, unposed). The 2D metric scores a query's mask volume over the eight views of a bundle. It ranks the first two changes in the opposite order from the 3D metric and does not see the third.

Multi-view features. Feeding the decoder with tokens computed one view at a time lowers 3D AP_{50} by 24% in the class-aware column and by half in the class-agnostic one. The shared queries rely on the cross-view information that the frozen backbone has already put into its tokens. This row is matched in schedule, not in convergence, since both runs were still improving at the last epoch.

3D anchors. Replacing 2D anchor boxes by 3D anchors is the clearest case of disagreement between the two metrics. In 2D it is neutral: 52.5 to 52.7 bundle AP_{50}, within the seed spread, with a small loss per frame. In 3D it raises AP_{50} from 6.7 to 11.2 (+67%), and by a similar factor under the posed transfer (15.6 to 25.7). The effect replicates: across the two seeds the class-agnostic gain is 8.8 and 7.0 points, eight to ten times the seed spread. The anchored model keeps 9% fewer queries per scene and covers 16% more mesh vertices, which suggests fewer duplicated or fragmented hypotheses, the kind of error that a per-vertex vote punishes.

Bundle width. Training with bundles of 16 instead of 8 views raises 3D AP_{50} by 46%, at twice the training time and memory. It does not combine with 3D anchors: under the unposed transfer the model with both is tied with the anchors-only model (13.9 against 13.8 class-agnostic, 10.9 against 11.2 class-aware), and under the posed transfer it is lower (23.6 against 25.7, Table 3). It keeps the fewest queries of all models and covers fewer vertices than the anchors-only one, a sign of over-pruning.

Consequence. A 2D metric, even one that scores a multi-view mask volume, is a poor proxy for the 3D benchmark. It ranks the first two rows of Figure 3 in the opposite order and is blind to the third. Any mechanism aimed at cross-view behaviour should be judged on the 3D metric.

Table 4: Single-variable ablations of the cross-view mechanisms, all trained on the official ScanNet split and evaluated on its validation set. The reference uses shared queries, multi-view backbone features, cross-frame attention, 2D anchor boxes and bundles of S=8 views; every other row differs from it in one training flag. The 2D columns are our own per-view mask metric on the 37×37 grid (AP_{50}; "bundle" scores one query's mask volume over 8 views) and exist for model selection only; the 3D columns are the official evaluator under the unposed transfer at 17.4 views. Seed-to-seed spread is about 0.9 points on both the 2D bundle AP_{50} and the 3D class-agnostic AP_{50}.

	2D AP_{50} (internal)		3D, class-aware			3D, class-agnostic		
Configuration	frame	bundle	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
Reference	65.0	52.5	2.3	6.7	26.8	1.3	5.0	32.0
second seed	65.1	53.4	2.5	7.5	31.3	1.6	5.9	34.8
without cross-frame attention	58.8	38.9	1.0	2.9	16.7	0.5	2.1	21.4
with per-frame backbone features	n/a	35.9	2.0	5.1	25.1	0.7	2.5	23.4
with 3D anchors	64.1	52.7	3.8	11.2	36.0	4.2	13.8	50.4
second seed	64.7	53.6	3.7	11.2	34.2	3.9	12.9	48.5
with wider bundles, S=16	66.2	55.2	3.3	9.8	33.6	2.3	8.0	39.1
with 3D anchors and S=16	64.6	53.6	3.2	10.9	35.3	4.1	13.9	50.4

Table 5: Cross-view identity of the shared queries, scored with tracking metrics by reading the 8 views of a bundle as time steps and each query as a track (ScanNet val). ID-sw. is our own diagnostic: the fraction of visible views in which another query fits a matched instance better than its own (lower is better).

	HOTA	AssA	DetA	IDF1	ID-sw.
2D anchors, seed 0	41.83	57.91	31.30	48.41	49.82
2D anchors, seed 1	40.80	58.37	29.66	46.34	47.10
3D anchors, seed 0	42.17	58.37	31.44	49.21	40.99
3D anchors, seed 1	41.04	58.12	29.96	47.72	40.74
Effect of 3D anchors	+0.29	+0.11	+0.22	+1.09	−7.59
Largest seed spread	1.12	0.46	1.65	2.08	2.72

5.5 Does a shared query keep its identity?

A model with view-shared queries can be scored as a tracker without any extra machinery: the views of a bundle are the time steps and each query is a track. Table 5 reports HOTA and its association and detection components [35] and IDF1 [47]. The shared queries reach an association accuracy of about 58, a detection accuracy of 30 to 31 and a HOTA of 41 to 42.

The table also corrects a claim we made during the project. Our own diagnostic, the fraction of views in which some other query fits a matched instance better than its assigned one, drops from 0.50 to 0.41 with 3D anchors, and we had read this as evidence that 3D anchors improve cross-view identity. None of the four standard metrics supports that reading: each moves by less than its own seed spread. The diagnostic is a winner-take-all statistic and flips on near ties between queries that segment the same object almost equally well, which the association accuracy does not reward. The 3D gain of Section 5.4 is measured on the benchmark and is unaffected, but its explanation is not "better association"; the query and coverage statistics above point to fewer competing hypotheses per object instead.

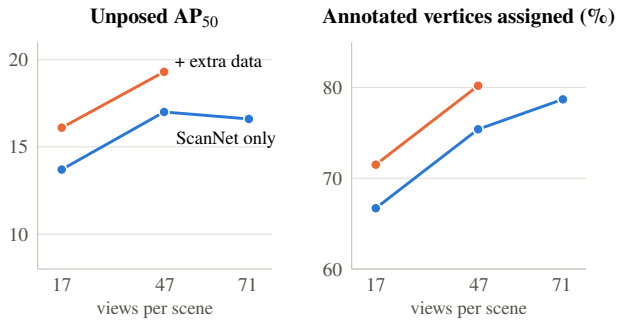


Figure 4: More views help up to about 50. Left: unposed class-agnostic AP_{50} on ScanNetv2 as a function of the mean number of views per scene. Right: fraction of annotated mesh vertices that end up in a predicted instance. Coverage keeps growing while AP_{50} does not. Blue: trained on ScanNet. Orange: with ScanNet++ and Infinigen added.

5.6 Lifting: views and registration

View budget. Going from 17 to 47 views raises unposed AP_{50} from 13.7 to 17.0 for the ScanNet-only model and from 16.1 to 19.3 with extra data (Figure 4, Table 14). Going further, to 71 views, changes nothing (16.6), although the fraction of annotated vertices covered continues to rise from 75% to 79%. The budget at which the published unposed methods report is thus also the one at which our result saturates, and coverage is not what limits it beyond that point.

Vote radius and registration. After the similarity alignment, the median error of the predicted camera centres on ScanNet is 0.14 m, and the residual of the ICP refinement about 0.10 m, both larger than the 5 cm radius within which a point may vote. Widening the radius helps until about 15 cm and not beyond (Table 13): AP_{50} rises from 6.7 to 9.0 for the reference model and from 13.8 to 18.5 (class-agnostic) for the 3D-anchor model. For the latter every radius and confidence setting in the sweep is at least as good as the default. Because the sweep was run on the evaluation set, we treat it as a sensitivity analysis and report the default everywhere else. These two parameters alone

Table 6: Training data. One recipe (one-class head, shared queries, 3D anchors, S=8), different training mixtures; all rows are evaluated on ScanNetv2 val, class-agnostic, at 17.4 views. SN = ScanNet, SN++ = ScanNet++, Inf = Infinigen. RE10K is capped at 1,500 scenes and ASE at 1,000; RE10K masks are SAM 2 pseudo-labels. The top block uses a learning rate of 10^{-4}, the bottom block $5{\times}10^{-5}$ and the final checkpoint (see text).

Training sources							Unposed			Posed		
SN	SN++	Inf	RE10K	ASE	Scenes	Steps	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
✓					1,201	42k	4.1	13.0	47.4	13.9	32.3	66.9
✓	✓				2,054	41k	4.2	12.9	48.0	14.0	32.7	66.8
✓	✓	✓			3,520	42k	3.2	10.6	45.0	11.1	27.7	64.1
✓	✓	✓			3,520	84k	5.7	16.6	51.6	17.7	38.9	70.8
	✓	✓			2,319	83k	0.3	1.3	21.2	0.8	2.9	32.3
	✓	✓	✓		3,819	84k	0.5	2.3	25.1	1.8	6.3	39.9
	✓	✓	✓	✓	4,819	87k	0.9	3.2	30.1	3.2	10.6	47.9

move the result by an amount comparable to the bundle-width change of Section 5.4.

Reading. At default lifting settings, every unposed result on ScanNet has an AP_{25} of roughly three times its AP_{50} or more (2.9 at the lowest), and more than nine times for the zero-shot model: instances are found and placed, but the lifted masks rarely overlap the ground truth tightly enough. Together with the factor of 2.3 of Section 5.3, the saturation in views and the scale of the registration error, this points to the alignment between predicted geometry and the benchmark mesh as the limiting step of the unposed pipeline.

5.7 Training data

Table 6 varies the training mixture under one recipe. The sources and their preparation are described in Appendix D.

Adding data to ScanNet. ScanNet++ is neutral on the ScanNet validation set (13.0 and 12.9 AP_{50}) and pays on other benchmarks (Section 5.8). Adding the synthetic Infinigen scenes first appears to hurt (10.6) and, with the number of steps doubled, gives the best model we trained (16.6). The first reading was an artefact of the step budget: at equal steps, each ScanNet scene is seen a third as often in the three-source mixture. The opposite confound then remains. The 84k-step model has twice the steps of the two smaller ones, and the control that would separate data from compute, a two-source model at 84k steps, became unstable at our default learning rate (Appendix D). The ScanNet-only model had stopped improving well before its last epoch, so more steps alone are unlikely to explain the gain, but we have no valid measurement of it.

Removing ScanNet. Without ScanNet the model is much weaker on ScanNet, and every added source helps: from 1.3 AP_{50} with ScanNet++ and Infinigen, to 2.3 with RE10K, to 3.2 with ASE. The 1,000 ASE scenes raise unposed AP_{50} by a factor of 1.4 and posed AP_{50} by 1.7.

The value of a source depends on the mixture. Table 7 isolates RE10K, whose masks are SAM 2 pseudo-labels [45] rather than annotations. Added to a mixture that contains ScanNet, at equal steps, it lowers ScanNet AP_{50} by 42%; added to one that does not, it raises it by a factor of 1.8. The likely reading is displacement: where in-domain real data is present, a redundant source takes optimisation steps away from it, and where it is absent, the same source is the closest available proxy. A statement such as "dataset X helps" therefore needs the mixture it was measured in.

Table 7: The value of a data source depends on what it is added to. Adding 1,500 RE10K scenes (SAM 2 pseudo-labels) to a mixture with or without ScanNet, at matched gradient steps and learning rate (SN = ScanNet, SN++ = ScanNet++, Inf = Infinigen). ScanNetv2 val, class-agnostic AP_{50}, 17.4 views.

Base mixture	Transfer	Base	+ RE10K	Change
SN, SN++, Inf	unposed	15.5	9.0	−42%
	posed	36.0	26.4	−27%
SN++, Inf	unposed	1.3	2.3	×1.8
	posed	2.9	6.3	×2.1

5.8 Other benchmarks

Table 8 applies the same models, evaluator and transfers to ScanNet200 [48], ScanNet++ [62] and Replica [51], class-agnostic because our label set does not cover their taxonomies.

Unposed transfer fails out of domain. On ScanNet++ and Replica every one of our models scores 0.0 to 0.1 AP_{50} under the unposed transfer, whatever it was trained on, while FAST3DIS reports 3.5 and 13.8 and the re-evaluated PanSt3R 6.2 on both. This is the weakest result of the study. The posed transfer shows that the masks are not the only cause: with ground-truth geometry the same models reach up to 6.8 AP_{50} on ScanNet++ and 11.9 on Replica.

Two properties of our pipeline compound here. First, the registration degrades. The camera-centre error after alignment grows from 0.14 m on ScanNet to 0.20 m on ScanNet++ and 0.28 m on Replica, where only 66% of points are ICP inliers. These two quantities depend only on the frozen geometry heads and on our alignment procedure, so no amount of head training changes them. For the ScanNet-trained 3D-anchor model of Table 4, the share of annotated vertices that receive an instance falls accordingly from 68% to 22% and 26%, against 69% under the posed transfer (Table 15). Second, neither dataset

Table 8: Other benchmarks, class-agnostic, same evaluator and the same two transfers. ScanNet200 re-labels the ScanNet validation scans with 200 categories (17.4 views); ScanNet++ (49 scenes) and Replica (8 scenes) use 50 views. Rows of ours share one recipe and differ in the training mixture. ‡The training data contains ScanNet++ scenes, so the ScanNet++ column is not zero-shot; our models exclude the 49 evaluation scenes from training. All other ScanNet++ and Replica cells are zero-shot. *As re-evaluated by FAST3DIS [25]. The evaluation scenes and instance sets of the published rows are not specified, so they are a reference and not a matched comparison.

		ScanNet200			ScanNet++			Replica		
Transfer	Model (training data)	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
Unposed	IGGT* [28]‡				0.6	2.2	10.0	1.1	3.9	15.2
	FAST3DIS [25]				1.0	3.5	14.7	5.1	13.8	30.3
	PanSt3R* [69]‡				1.7	6.2	21.7	1.7	6.2	20.7
	Ours (ScanNet)	3.2	9.9	34.3	0.0	0.0	1.0	0.0	0.0	0.4
	Ours (ScanNet, ScanNet++)‡	3.3	9.9	34.5	0.0	0.0	1.6	0.0	0.0	1.1
	Ours (ScanNet, ScanNet++, Infinigen)‡	4.1	11.8	37.2	0.0	0.1	2.6	0.0	0.0	3.6
	Ours (no ScanNet)‡	1.7	5.3	27.3	0.0	0.0	0.7	0.0	0.1	1.8
Posed	Ours (ScanNet)	10.7	24.1	49.6	0.6	2.7	14.3	1.4	4.7	21.1
	Ours (ScanNet, ScanNet++)‡	10.9	24.3	50.2	1.2	4.3	23.4	3.2	8.0	32.6
	Ours (ScanNet, ScanNet++, Infinigen)‡	13.2	28.7	53.9	1.9	6.8	27.5	4.0	11.9	48.0
	Ours (no ScanNet)‡	6.1	14.9	44.5	0.5	1.9	18.2	2.1	8.2	37.1

provides an over-segmentation that we could use, so the vote is per vertex on meshes with about a million vertices and sparse votes are not propagated as they are on ScanNet. Our lifting was designed and left at its defaults on ScanNet, and the published rows use their own. The comparison therefore says that our unposed pipeline does not transfer, not that frozen VGGT geometry cannot.

Training data helps where the transfer allows it. Under the posed transfer the effect of data is large. Adding ScanNet++ leaves ScanNet200 unchanged and raises AP_{50} from 2.7 to 4.3 on held-out ScanNet++ scenes and from 4.7 to 8.0 on Replica, which none of the models has seen. The three-source model reaches 6.8 and 11.9. The model trained without ScanNet matches the two-source model on Replica (8.2 against 8.0), and ASE accounts for most of that: it multiplies Replica AP_{50} by 3.6 where it multiplied ScanNet AP_{50} by 1.4 to 1.7 (Table 10). As with ScanNet++ above, the gain is largest on the benchmarks furthest from ScanNet.

6 Discussion and Limitations

What a frozen backbone buys and costs. Freezing the backbone makes the recipe cheap: about four GPU-hours of optimisation for the ScanNet model and about nineteen hours for an 84k-step mixture, against the sixteen GPU-days reported by SegVGGT and IGGT. It also leaves the geometry outputs of the backbone exactly as released. The price shows in the two comparisons where the training data is aligned. Against SegVGGT, on the same split and under its own transfer, the frozen model is 2.8 times lower at AP_{50}. Against FAST3DIS and IGGT, without ScanNet, it is 3.0 and 3.5 times lower at AP_{50} and level at AP_{25}; PanSt3R, frozen as well, sits at the same AP_{50}. The pattern has one direction in both comparisons: the distance is largest at the strict overlap threshold and narrows at the coarse one, to a factor of 1.5 to 1.7 against SegVGGT and to parity in the zero-shot case. A decoder on frozen features finds and coarsely places instances far better than it delineates them.

What cannot be concluded. We did not train an adapted variant of our own model, so the factor of three is a distance between systems, not the measured effect of freezing. In every comparison the systems also differ in training scale, view budget, mask resolution and query design, and PanSt3R's authors report that fine-tuning their backbone changed little for panoptic quality. The experiment that would turn this distance into a cost is a minimal adaptation of the same model (for instance LoRA of low rank) under the protocol of this paper. Our 2D measurements suggest that mask resolution is not the first thing to fix: ground-truth masks quantised to the 37×37 grid still score 95.6 AP_{50} at full image resolution in a 2D check (Appendix B). The lifting results of Sections 5.6 and 5.8 point at registration instead.

Limitations. Most cells are single runs; two seeds exist only for the reference and 3D-anchor models at 17.4 views, and the 47-view rows are single runs. Checkpoints are selected on the ScanNet validation split by a 2D metric, which is common practice but not a held-out selection. The ScanNet training frames come from the first tenth of each scan, while the evaluation frames span the whole scan. The zero-shot model was scored at 17.4 views only. All competitor numbers are taken from the papers; we re-ran none of them, and two are themselves re-evaluations by a third party. The lifting procedure is simple and was not adapted to benchmarks without an over-segmentation, which is part of the reason for the out-of-domain failure of Section 5.8. The class head covers 19 ScanNet categories, so class-aware results exist only on ScanNetv2. One of the training sources is pseudo-labelled. We report no qualitative results and no inference timing, and we study one backbone.

Recommendations for reporting. The audit suggests four items that a paper on image-based 3D instance segmentation should state with every benchmark number: how masks are transferred to the point cloud (predicted or ground-truth geometry), the label setting, the number of input views, and whether the benchmark's training split was used. Where possible both transfers should be reported, since their ratio separates segmentation from reconstruction. Mechanisms that act across views should be evaluated on the 3D metric, not on a 2D proxy.

7 Conclusion

We trained a query decoder on a strictly frozen VGGT for multi-view 3D instance segmentation and measured it under the protocols of the methods that adapt their backbone. Three results stand out. First, those protocols are not interchangeable: the choice between predicted and ground-truth geometry in the mask transfer is worth a factor of 2.3 in AP_{50}, and the training domain is worth more. Second, once the training data is aligned, the frozen model is about three times lower at AP_{50} than adapted ones, whether the reference is a method trained on the same split or methods evaluated zero-shot, and the distance narrows at AP_{25}, to parity in the zero-shot case. Its in-domain lead over zero-shot methods is a data effect. Third, the parts of the design that matter are the ones that connect views, and the step that limits the result is the lifting to 3D; a 2D mask metric ranks the first incorrectly and cannot see the second.

Two directions follow. One is the lifting through predicted geometry, above all its registration to the target frame, which caps the unposed result in domain and, together with a vote that has no over-segmentation to rely on, breaks it out of domain. The other is the smallest adaptation of the backbone that closes part of the distance at the strict threshold, measured under matched protocols, which would replace the distance reported here by a cost curve.

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Table 9: Implementation details. Values shared by every run unless a table says otherwise.

Backbone (frozen)	
Model	VGGT-1B, last aggregator layer, 2048-d tokens
Input	518×518 (resized without cropping), patch 14
Token grid	37×37 per view; camera and register tokens dropped
Trainable head	
Pixel decoder	3 levels (37^2, 19^2, 10^2), 6 deformable encoder layers, 256-d, 8 heads, 4 points
Decoder	9 layers, 300 queries, 256-d, 8 heads, 4 points per level
Query init.	two-stage selection; mask-derived initial boxes
Denoising	100 queries, noise scale 0.4
Classes	19 sigmoid logits, or 1 (class-agnostic)
Mask resolution	37×37
Parameters	22.8 M (23.0 M with 3D anchors); 20.5 M without cross-frame attention
Optimisation	
Loss weights	class 4, mask 5, Dice 5, box L_1 5, GIoU 2
Optimiser	AdamW, weight decay 0.05, gradient clip 0.1
Learning rate	10^{-4} ($5{\times}10^{-5}$ for the lower block of Table 6 and for Table 7)
Schedule	2 warm-up epochs, cosine decay to 5%
Batch	one bundle of S=8 views per step
ScanNet recipe	12 epochs, 2 bundles per scene, 28,824 steps
Augmentation	brightness and contrast jitter (0.2) on the second bundle
3D evaluation	
Queries kept	top 100 by class score
Mask threshold	0.5 on bilinearly upsampled logits
Registration	Sim(3) from camera centres, 10 ICP iterations
Vote radius	5 cm; no depth-confidence filtering
Instances	plurality vote per superpoint

A Implementation Details

Table 9 lists the settings shared by all runs.

Decoder port. The pixel decoder and the query decoder are a port of the reference Mask DINO implementation in plain PyTorch, with the multi-scale deformable attention computed by bilinear sampling instead of the compiled operator. As a check of the inference path, loading the released COCO weights into the ported modules reproduces the published COCO mask AP to within 0.01 (46.133 against 46.129). Differences from the reference recipe are limited to what the setting requires: the backbone and its single-scale feature map, the class head (19 sigmoid logits, or one), masks supervised on the 37×37 token grid, and no geometric augmentation, since the cached tokens fix the image geometry.

Feature cache. Tokens are stored in half precision, about 45 MB per bundle of eight views, and held in host memory during training together with the ground truth. The training process peaks at 135 GB for the ScanNet training and validation splits at one bundle per scene, and at 258 GiB for the 3,520-scene mixture. Memory, not GPU time, is what bounds the size of the training set in this design. With 3D anchors, the per-token positions add 0.15% to the cache.

Cross-frame attention. The block is a multi-head self-attention layer (8 heads, 256 dimensions) with a residual connection and layer normalisation. Its input sequence is the S per-view copies of one query, and the batch is formed by all queries of all bundles. One block follows each of the nine decoder layers, before the anchor refinement of that layer, which adds 2.4 M parameters in total.

3D anchors. Token positions are the confidence-weighted means of the point-head output over each 14×14 patch. The centre and scale used to normalise a bundle are estimated from the tokens at or above the median confidence, so that unreliable points do not set the scale. The coarser pyramid levels inherit positions by nearest-neighbour resampling, which keeps every position on a surface. The radius is initialised to 0.25 in normalised units. Unlike a box refinement, the anchor update is not detached between layers, because the anchor has no loss of its own and learns only through the projection of Equation (1). Two-stage selection still produces mask-derived initial boxes, which supervise the initial prediction, but from the first decoder layer onward the projected anchor is the reference; the comparison in Table 4 therefore exchanges the box and its refinement for the anchor and its refinement as one unit. 3D anchors increase training time by 15% and require multi-view features, since per-frame point maps do not share a coordinate frame.

Schedules and cost. The ScanNet models of Table 4 train for 12 epochs with two bundles per scene (28,824 steps): 204 minutes of optimisation for the reference model and 233 with 3D anchors on one RTX 4090. The complete job of the reference model, which also fills the cache and evaluates on the validation split after every epoch, took 5.7 hours; the same job with bundles of 16 views took 11.4 hours on an A100 with 80 GB. The mixture models of Table 6 use one bundle per scene and the step counts listed there; the 84k-step run on 3,520 scenes takes 18.8 hours on one A100. No duration was recorded for the models trained without ScanNet. One 3D evaluation of the 312 validation scenes takes about 46 minutes at 17 views.

Inference. All sampled views of a scene form one bundle. Since neither the shared initialisation nor the cross-frame attention depends on the number of views, models trained with S=8 are applied to 17 to 71 views without modification.

B Evaluation Details

Evaluator. The 3D metrics come from a port of the official ScanNet instance evaluation script to Python 3 that operates on arrays instead of text files. The overlap thresholds, the minimum region size of 100 vertices, the confidence-ordered matching, the handling of duplicate detections and the treatment of unannotated vertices are unchanged. Per-vertex ground truth

Table 10: Full cross-dataset matrix, class-agnostic (AP / AP_{50} / AP_{25}). Every cell is one run at default lifting parameters; no scene failed in any cell. The first three rows are the class-aware checkpoints of Table 4 scored with collapsed labels; the others train a one-class head on the mixture shown (SN = ScanNet, SN++ = ScanNet++, Inf = Infinigen). Rows whose mixture contains SN++ are not zero-shot on the ScanNet++ column.

Model	Transfer	ScanNetv2	ScanNet200	ScanNet++	Replica
SN, 2D anchors	unposed	1.3 / 5.0 / 32.0	2.3 / 6.9 / 27.8	0.0 / 0.0 / 0.4	0.0 / 0.0 / 0.7
	posed	3.9 / 12.2 / 48.3	7.5 / 17.1 / 41.1	0.3 / 0.9 / 7.5	0.4 / 2.3 / 9.6
SN, 3D anchors	unposed	4.2 / 13.8 / 50.4	3.6 / 11.1 / 36.6	0.0 / 0.0 / 1.5	0.0 / 0.0 / 0.9
	posed	10.9 / 30.4 / 67.7	12.4 / 27.5 / 52.3	0.9 / 3.8 / 17.8	0.6 / 2.8 / 19.0
SN, 2D anchors, S=16	unposed	2.3 / 8.0 / 39.1	2.7 / 8.4 / 31.4	0.0 / 0.0 / 1.3	0.0 / 0.0 / 0.8
	posed	6.4 / 19.0 / 57.2	9.9 / 22.4 / 47.4	0.7 / 2.1 / 10.8	0.8 / 4.6 / 12.5
SN (42k steps)	unposed	4.1 / 13.0 / 47.4	3.2 / 9.9 / 34.3	0.0 / 0.0 / 1.0	0.0 / 0.0 / 0.4
	posed	13.9 / 32.3 / 66.9	10.7 / 24.1 / 49.6	0.6 / 2.7 / 14.3	1.4 / 4.7 / 21.1
SN + SN++ (41k)	unposed	4.2 / 12.9 / 48.0	3.3 / 9.9 / 34.5	0.0 / 0.0 / 1.6	0.0 / 0.0 / 1.1
	posed	14.0 / 32.7 / 66.8	10.9 / 24.3 / 50.2	1.2 / 4.3 / 23.4	3.2 / 8.0 / 32.6
SN + SN++ + Inf (42k)	unposed	3.2 / 10.6 / 45.0	3.0 / 9.0 / 34.2	0.0 / 0.0 / 1.1	0.0 / 0.0 / 1.4
	posed	11.1 / 27.7 / 64.1	9.7 / 22.0 / 50.0	0.7 / 2.9 / 23.2	2.4 / 9.0 / 37.8
SN + SN++ + Inf (84k)	unposed	5.7 / 16.6 / 51.6	4.1 / 11.8 / 37.2	0.0 / 0.1 / 2.6	0.0 / 0.0 / 3.6
	posed	17.7 / 38.9 / 70.8	13.2 / 28.7 / 53.9	1.9 / 6.8 / 27.5	4.0 / 11.9 / 48.0
SN++ + Inf (83k)	unposed	0.3 / 1.3 / 21.2	0.5 / 2.1 / 18.3	0.0 / 0.0 / 0.1	0.0 / 0.0 / 0.5
	posed	0.8 / 2.9 / 32.3	1.5 / 4.9 / 28.7	0.0 / 0.1 / 9.6	0.3 / 1.6 / 19.4
SN++ + Inf + RE10K (84k)	unposed	0.5 / 2.3 / 25.1	0.9 / 3.3 / 22.3	0.0 / 0.0 / 0.3	0.0 / 0.1 / 0.6
	posed	1.8 / 6.3 / 39.9	3.0 / 8.6 / 36.4	0.1 / 0.6 / 12.8	0.5 / 2.3 / 27.8
SN++ + Inf + RE10K + ASE (87k)	unposed	0.9 / 3.2 / 30.1	1.7 / 5.3 / 27.3	0.0 / 0.0 / 0.7	0.0 / 0.1 / 1.8
	posed	3.2 / 10.6 / 47.9	6.1 / 14.9 / 44.5	0.5 / 1.9 / 18.2	2.1 / 8.2 / 37.1

is built from the official annotation files in the same way as the benchmark export. Besides unit tests on hand-computed cases, the ground truth of every evaluation scene, fed back as predictions, scores exactly 100 / 100 / 100 on each of the four benchmarks used in this paper.

Unposed transfer. The similarity transform is first solved in closed form from the predicted and ground-truth camera centres [53] and then refined by ten iterations of ICP [2] between the points unprojected from all pixels, not only the masked ones, and the mesh vertices, with a correspondence threshold of 0.3 m and the scale re-estimated at every iteration. The registration therefore does not depend on the masks. Only pixels that belong to a kept query cast votes. Optionally, the lowest-confidence fraction of depth pixels is discarded; the default is to keep all of them.

Posed transfer. Mesh vertices are transformed into each camera with the ground-truth pose, projected with the depth intrinsics into the sensor depth map, and kept if the projected depth agrees with the sensor reading within 0.1 m, the tolerance used by SegVGGT. Surviving vertices are projected into our mask grid. Because our inputs are resized to a square without cropping, this projection scales the two image axes differently, a detail that an isotropic rescaling would get wrong by about 40 pixel rows. The oracle of Section 4.2 exists to catch such errors: a wrong mapping lowers its purity far below the measured 99.99% (worst scene 99.77%).

Label setting. Table 12 scores identical predictions with and without class labels. Class-agnostic scoring is not uniformly easier. It lowers AP_{50} for the models with 2D anchors and raises it for the model with 3D anchors, in both seeds. The class-aware mean is carried by a few distinctive classes (on the reference model with a 10 cm vote radius and 25% of the depth discarded, toilet reaches 50.8 AP_{50}, sink and refrigerator 17.3) while frequent classes are weak (chair 5.3, cabinet 4.0) and the class our head cannot predict contributes zero. Pooling all instances into one ranking removes the leverage of rare classes and exposes duplicated detections of frequent ones. A model with fewer competing hypotheses per object gains from pooling; one with more loses. The sign of the difference is therefore a property of the model, which is one more reason not to compare a class-aware number with a class-agnostic one.

Internal 2D metrics. The per-frame metric scores each view separately against that view's ground-truth masks on the 37×37 grid and averages over frames and scenes. The per-bundle metric treats the masks of a query in the eight views of a bundle as one volume and computes one IoU against the ground-truth volume. In both, a query with no mask pixels in a view is dropped from that view rather than counted as a false positive, and queries are kept if their class score exceeds 0.25. For the tracking metrics of Table 5, the views of a bundle are taken as time steps, the thresholded queries as detections, and the query index as the track identity; HOTA is averaged over localisation thresholds from 0.05 to 0.95 as in the reference implementation.

Mask resolution. Whether the 37×37 mask grid limits the 2D result was checked before the official split was adopted,

Table 11: Datasets. Top: 2D instance supervision used for training. Bottom: 3D benchmarks used for evaluation. "Instances" counts evaluated ground-truth instances.

Training sources			
Source	Scenes	Annotation	Notes
ScanNet v2 [10]	1,201	projected 3D annotation	official training split; 17,638 instances; 19 classes
ScanNet++ [62]	853	per-frame instance ids	from the InsScene-15K release [28]; the 50 scenes of our ScanNet++ benchmark are removed
Infinigen [44]	1,466	rendered	from the InsScene-15K release; room shell (wall, floor, ceiling) removed by name
RE10K [66]	1,500	SAM 2 pseudo-labels	from the InsScene-15K release (5,127 scenes available); masks covering over 30% of the frame removed
ASE [1]	1,000	rendered	scenes 0 to 999 of the public release; 31,897 frames, 78,347 instances
Evaluation benchmarks			
Benchmark	Scenes	Instances	Notes
ScanNetv2 val [10]	312	4,364	18 classes; class-aware and class-agnostic
ScanNet200 val [48]	312	10,045	same scans, 200 categories; class-agnostic only
ScanNet++ val [62]	49	2,579	84 instance classes; 50 views per scene; class-agnostic only; one scene of the official 50 dropped for a corrupt trajectory
Replica [51]	8	368	50 views per scene; class-agnostic only; room shell and unlabelled objects removed from the instance set

Table 12: Label setting on identical predictions (ScanNetv2 val, unposed, 17.4 views). Class-agnostic scoring collapses all labels on both sides and replaces the mean over 18 classes by one instance-pooled ranking.

	Class-aware			Class-agnostic		
Checkpoint	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
2D anchors, S=8	2.3	6.7	26.8	1.3	5.0	32.0
second seed	2.5	7.5	31.3	1.6	5.9	34.8
3D anchors, S=8	3.8	11.2	36.0	4.2	13.8	50.4
second seed	3.7	11.2	34.2	3.9	12.9	48.5
2D anchors, S=16	3.3	9.8	33.6	2.3	8.0	39.1
trained longer	3.2	11.5	41.4	2.9	10.4	45.8

on a 490-scene subset of ScanNet with ten validation scenes; we report it as a preliminary measurement. Ground-truth masks downsampled to the grid and scored at the full 518×518 resolution reach 95.6 AP_{50} (99.2 at 74×74), far above the model's per-frame AP_{50}, and a model trained with 74×74 masks was not better than one at 37×37. On the official split, scoring a single-frame model at full resolution instead of on the grid changes per-frame AP_{50} from 66.2 to 65.1.

Other benchmarks. ScanNet200 uses the ScanNet validation scans with the 200-category annotation, which more than doubles the number of evaluated instances and makes wall and floor valid instances. For ScanNet++ we use the 50 scenes of the official validation list with 50 frames sampled uniformly from the phone video; one scene is dropped because its released camera trajectory is corrupt. For Replica we use the eight commonly used scenes with 50 frames each, and an instance set from which wall, floor, ceiling and unlabelled objects are removed, in line with the other two benchmarks; Replica numbers depend on this choice. The ScanNet++ release contains no over-segmentation and the one in Replica mixes objects, so the vote is per vertex on both. On every dataset, unprojected sensor depth lands within 2 cm of the mesh in the median scene, which confirms the pose and depth conventions of our loaders.

C Additional Results

Lifting parameters. Table 13 gives the sweep summarised in Section 5.6. The vote radius saturates near 15 cm for both models. Discarding low-confidence depth helps the reference model slightly and does not help the 3D-anchor model, so a value tuned on one checkpoint should not be carried to another. Keeping 600 queries instead of 100 has no effect.

View budget. Table 14 gives the numbers behind Figure 4, together with the posed result at 47 views. The 17-view rows here use frames extracted from the raw scans at the same sampling as the standard export, so that only the number of views varies within the table; the standard export re-compresses its images, which explains the small differences from the 17.4-view rows of Table 2 (for instance 13.7 against 13.8 AP_{50}).

Full matrix. Table 10 lists every model on every benchmark under both transfers. Three observations complement Section 5.8. The ranking of the three class-aware checkpoints is the same on ScanNet200 as on ScanNetv2 under the posed transfer, with 3D anchors first. The registration statistics of the unposed transfer are the same for every model on a given set of frames, since they depend on the frozen backbone alone. And on rows with near-zero values the ratios between models are not meaningful; we quote them only where both numbers are above one point.

Diagnostics. Table 15 supports the reading of Section 5.8. With ground-truth geometry, the out-of-domain scenes are covered about as well as ScanNet scenes are under the unposed transfer; with predicted geometry, coverage falls to about a

Table 13: Sensitivity to the two lifting parameters (ScanNetv2 val, unposed, 17.4 views): the radius within which an unprojected pixel may vote for a mesh vertex, and the percentage of lowest-confidence depth pixels discarded. The sweep is run on the evaluation set, so it is a sensitivity analysis; all other tables use the first row of each block.

		Class-aware			Class-agnostic		
Radius	Drop	AP	AP_{50}	AP_{25}	AP	AP_{50}	AP_{25}
2D anchors							
5 cm	0%	2.3	6.7	26.8	1.3	5.0	32.0
5 cm	25%	2.3	7.1	26.0			
10 cm	0%	2.6	7.8	32.1			
10 cm	25%	2.9	8.3	30.5	1.7	6.0	33.4
10 cm	50%	2.4	6.8	26.9			
15 cm	25%	3.0	9.0	32.1			
20 cm	25%	2.9	9.0	32.5			
30 cm	25%	2.9	9.1	32.6			
3D anchors							
5 cm	0%	3.8	11.2	36.0	4.2	13.8	50.4
10 cm	0%	4.7	14.2	39.5	5.2	17.1	54.5
15 cm	0%	4.8	15.1	41.9	5.5	18.5	57.1
10 cm	25%	4.7	13.6	38.1	5.1	16.8	51.8
15 cm	25%	5.0	15.1	39.6	5.5	18.0	54.4
20 cm	25%	5.0	15.1	40.3	5.5	17.8	54.8
3D anchors, 600 instead of 100 kept queries							
5 cm	0%	3.8	11.1	35.7	4.3	14.0	50.2

quarter of the annotated vertices.

D Training Data

Table 11 lists the training sources and the benchmarks.

Preparation. For ScanNet we use the official 2D instance and label images of the training split. ScanNet++, Infinigen and RE10K are taken from the public release of InsScene-15K [28], which provides per-frame instance maps with identities that are consistent within a scene; we verified this property on each source before use. ASE is not part of that release and was obtained from its own distribution [1], which includes rendered instance maps. From every scene we keep up to 32 evenly spaced frames, resized to 518×518. Since these sources do not share a label set, all mixture models are trained with a single object class.

Exclusions. The 50 scenes of the ScanNet++ validation list are removed from the ScanNet++ training source, so that our ScanNet++ benchmark contains no training scene. In Infinigen, walls, floors and ceilings are removed by name, because none of the benchmarks treats the room shell as an instance; in ScanNet++ no instance dominates a frame and no filter is applied. ASE needs none either: in a sample of its scenes the largest instance covers at most 32% of a frame.

Pseudo-labels. The RE10K masks in InsScene-15K are produced by SAM 2 [45], not annotated. They are a class-agnostic over-segmentation of the whole image, room shell included. We drop masks that cover more than 30% of the frame on

Table 14: View budget (ScanNetv2 val, class-agnostic). Views are the mean number of frames per scene when 17, 50 and 100 are requested. "Voted" is the fraction of mesh vertices that receive a vote, "assigned" the fraction of annotated vertices that end up in a predicted instance (unposed transfer).

		Unposed			Posed	Coverage	
Training data	Views	AP	AP_{50}	AP_{25}	AP_{50}	voted	assigned
ScanNet	17.0	4.4	13.7	48.8		17.1	66.7
	46.7	5.3	17.0	54.2	33.6	26.8	75.4
	71.3	5.2	16.6	53.6		30.8	78.7
+ extra data	17.0	5.6	16.1	50.3		24.1	71.5
	46.7	6.9	19.3	56.0	41.9	36.8	80.2

Table 15: Coverage and registration by dataset for the ScanNet-trained 3D-anchor model. "Assigned" is the median fraction of annotated vertices that receive a predicted instance; the last two columns describe the alignment of the unposed transfer.

	Assigned (%)		ICP	Camera
Dataset	posed	unposed	inliers (%)	error (m)
ScanNetv2	83.4	67.9	96.3	0.136
ScanNet++	68.5	22.3	92.4	0.200
Replica	68.5	25.5	66.0	0.275

average, which together with a minimum-area filter removes 1.8% of the masklets; parts of walls and floors below that size remain as supervised instances. RE10K is capped at 1,500 of the 5,127 available scenes and ASE at 1,000 scenes, in both cases because of the host memory required by the feature cache. Every result that involves RE10K is marked as such.

Step budgets. One step is one bundle, so at a fixed step budget a larger mixture passes over each of its sources fewer times. At matched steps the three-source mixture looked worse than the smaller ones, and it came out best once trained longer (Section 5.7). Every row of Table 6 therefore states its step budget.

Learning rate. The default learning rate of 10^{-4} is stable up to the 3,520-scene mixture at 42k and 84k steps. It failed in two runs: the four-source mixture with ScanNet, and the two-source mixture trained for 84k steps. In both the training loss began to rise right after warm-up and accuracy on the training scenes themselves collapsed, which distinguishes the failure from slow convergence. Halving the rate removed it at unchanged data in the first case. All runs of the lower block of Table 6 and of Table 7 therefore use 5×10^{-5}, and every comparison is made between runs at the same rate. The failed runs are not used for any number in this paper.

Checkpoint selection without in-domain validation. For models trained without ScanNet, the ScanNet validation set is a zero-shot signal, and selecting on it proved unreliable: the 2D validation metric peaked in the first few epochs and fluctuated afterwards while accuracy on the training sources kept improving. For these models we report the final checkpoint.